Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

AI Engineer with 3 years of industry experience at HDFC Asset Management building agentic AI systems and RAG applications end to end: Azure OpenAI tool-calling agents, LlamaIndex RAG over PostgreSQL with pgvector, and FastAPI services with a ReactJS frontend, deployed on AWS in Docker through CI/CD. First engineer on the GenAI systems alongside my manager, taking them to real users; a WhatsApp fund-comparison agent is used daily by roughly 50 to 100 Sales team members. De facto team lead for up to 4 interns; Best Team Project twice. MSc in Data Science at the University of Bristol (awaiting result) with an AstraZeneca dissertation. Seeking an AI Engineer role building a proprietary agentic AI platform from the ground up.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Agentic AI & RAG Applications in Production

- Designed and built production **GenAI** applications end to end (**ReactJS**, **Python FastAPI** microservices, **LlamaIndex RAG**, **Azure OpenAI GPT-3.5 & Llama-3**) for Compliance, Client Service and Investment, initially as the first engineer on the team alongside my manager; a separate **pgvector** database with **Azure OpenAI** embeddings per department kept access siloed.
- Built an **agentic WhatsApp fund-comparison bot** (via the Netcore API) used daily by the Sales team (roughly 50 to 100 users, 400 to 1,000 requests in working hours): **Azure OpenAI tool calling** picks the comparison type and funds and calls **Python functions** over SQL data, returning fund indicators and a chart PDF. Parameter extraction broke first; fixed with example inputs in the JSON tool definitions.
- Took applications from prototype to real users, iterating on what broke; guardrailed behaviour through **system prompts**, versioned every prompt change in **Git** and had Compliance review each release; allocated models by cost, self-hosting **Llama-3** on a **GPU EC2** instance for the largest team.

Data Pipelines & Automation

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly and integrated with legacy analyst data; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.
- Built a **Python** pipeline that parses over **300 daily emails** from credit rating agencies, fuzzy-matches them to the companies analysts track (dropping the rest) and sends a daily summary of previous and changed ratings, **saving 60 analyst hours a month**; replaced **AWS Textract** with **Azure OpenAI** when an agency began sending images, which also improved accuracy.
- Developed **Power BI** dashboards for the investment team's credit report analysis, moving analysts from Excel and improving efficiency by **10 hours a week**; recreated them in the ReactJS frontend so the wider organisation could use the data without Power BI licensing costs.
- Worked directly with Investment, Compliance and Client Service stakeholders; **de facto team lead for up to 4 interns**; **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

MLOps, Deployment & Reliability

- Designed the infrastructure architecture for internal analytics platforms and GenAI applications on **AWS (EC2 including a GPU instance, S3, ECR and Lambda**), packaged the **FastAPI** microservices as **Docker** containers and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins and GitLab CI/CD** pipeline.
- Ran both **real-time and batch workloads**: user-facing services on FastAPI microservices, and a daily rating-email pipeline run on **AWS SageMaker** and scheduled by an **EventBridge** rule that triggers a **Lambda** function at 9 AM.
- Implemented **RBAC** with **LDAP** authentication, restricted access to company private IPs with **AWS Security Groups** and deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**.

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. My focus: the **RAG-based AI assistant** that narrates ranked results and answers methodology questions; the **scoring and ranking algorithm** across six evidence layers; and the **data preprocessing pipeline** (14 clean, joined tables from 17 raw inputs, with integrity checks). Returns a ranked shortlist of cancer cell lines with the evidence behind each pick, refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app with **108 LLM personas** across 12 archetypes and a vision model that scores uploaded photos (ReactJS, Python FastAPI, NVIDIA NIM with the model selectable per request). LLM output treated as untrusted: **validated with Pydantic, repaired and cached by input hash**; rate limits shaped the design (12 calls instead of 108, results streamed). GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. **Multi-architecture deep learning study** comparing CNN, ResNet and ViT encoders trained from scratch with LSTM/attention decoders on three synthetic datasets across five seeds, evaluated with SBERT cosine similarity, BLEU and ROUGE-L and benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git

ACHIEVEMENTS

- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle credit card fraud detection problem.)
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework for AI-assisted coding) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2 incl. GPU, S3, ECR, Lambda, SageMaker, EventBridge), Azure OpenAI, LlamaIndex, Docker, Jenkins, GitLab, Git, Nginx, Linux, LDAP, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite